Efficiency Of Second-Order Optimization: A Comparative Implementation Of The Gauss-Newton Algorithm For Non-Linear Model Fitting

PROJECT REPORT

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1. Abstract

The optimization of non-linear models remains a foundational challenge in machine learning and computational mathematics. While first-order methods such as Stochastic Gradient Descent (SGD) and Adam dominate deep learning due to their low computational cost per iteration, they often suffer from slow convergence rates in highly non-linear, poorly conditioned topologies. This report details the theoretical derivation, practical implementation, and empirical evaluation of the Gauss-Newton (GN) algorithm, a specialized second-order optimization technique designed specifically for non-linear least squares problems.

By avoiding the explicit computation of the full Hessian matrix, the Gauss-Newton method approximates second-order curvature, offering quadratic convergence near the optimal solution. This project implements a custom Gauss-Newton optimizer from scratch and rigorously compares its performance against standard optimization techniques using a benchmark regression dataset. The evaluation metrics—spanning Loss vs. Epochs, Loss vs. Time, Mean Squared Error (MSE), and the R^2 coefficient of determination—demonstrate the distinct trade-offs between step complexity and convergence speed. The complete system is encapsulated within a user-friendly graphical interface, serving as an end-to-end framework for mathematical optimization analysis.

2. Introduction

2.1 Problem Statement

In the realm of predictive modeling, non-linear regression aims to fit a mathematical function to a set of observations by minimizing a specific loss function, typically the Sum of Squared Residuals (SSR). As models grow in complexity, the loss landscape becomes increasingly non-convex, characterized by saddle points, flat regions, and steep ravines. Standard first-order optimizers (like Gradient Descent) navigate this space using only gradient information, frequently leading to oscillatory behavior or premature stagnation in local minima.

Second-order methods, which utilize both gradient (first derivative) and curvature (second derivative) information, provide much more accurate update steps. However, computing the full Hessian matrix for Newton's Method is computationally prohibitive for high-dimensional data ($O(N^3)$ complexity for matrix inversion). There is a critical need for algorithms that capture second-order curvature without the crippling computational overhead of full Hessian formulation.

2.2 Objectives of the Study

This project is driven by the following core objectives:

1. **Theoretical Formulation:** To mathematically derive the Gauss-Newton algorithm from first principles, proving its viability as a Hessian approximation technique.
2. **Custom Implementation:** To engineer a robust, numerically stable implementation of the Gauss-Newton optimizer within a modern deep learning framework (e.g., PyTorch).
3. **Empirical Benchmarking:** To systematically compare the convergence behavior of Gauss-Newton against traditional first-order methods over varying epochs and wall-clock time.
4. **System Integration:** To package the optimization logic, data loading, and evaluation scripts into a cohesive, production-ready pipeline complete with a Graphical User Interface (GUI).

2.3 Scope and Limitations

This report focuses on unconstrained non-linear least squares optimization. The custom Gauss-Newton implementation is evaluated on structured tabular data (train.csv, test.csv). While highly effective for regression tasks with hundreds to thousands of parameters, the algorithm's reliance on Jacobian matrix inversion limits its scalability to massive deep neural networks (millions of parameters), where iterative solvers like Conjugate Gradient or pseudo-second-order methods like L-BFGS are traditionally preferred.

3. Theoretical Framework and Mathematical Foundations

3.1 Unconstrained Optimization Basics

Let $f(x,\beta)$ be a non-linear model predicting a response y, parameterized by a weight vector $\beta=[\beta_1,\beta_2,\dots,\beta_n]^T$. Given a dataset of m observations (x_i, y_i), the objective is to minimize the residual sum of squares (RSS). We define the residual function for the i-th observation as:

$$r_i(\beta)=y_i-f(x_i,\beta)$$

The overall cost function to be minimized is the sum of squared residuals:

$$S(\beta)=\frac{1}{2}\sum_{i=1}^{m}(r_i(\beta))^2=\frac{1}{2}\|r(\beta)\|_2^2$$

Where $r(\beta)$ is the vector of residuals: $r(\beta)=[r_1(\beta),r_2(\beta),\dots,r_m(\beta)]^T$.

3.2 Gradient Descent and First-Order Methods

First-order methods update the parameters by moving in the direction of the negative gradient of the objective function:

$$\beta^{(k+1)}=\beta^{(k)}-\alpha\nabla S(\beta^{(k)})$$

Where α is the learning rate. The gradient of $S(\beta)$ can be expanded using the chain rule:

$$\nabla S(\beta)=\sum_{i=1}^{m} r_i(\beta)\nabla r_i(\beta)=J_r(\beta)^T r(\beta)$$

Here, $J_r(\beta)$ is the **Jacobian matrix** of the residuals with respect to the parameters β. It is an $m\times n$ matrix defined as:

$$(J_r)_{ij}=\frac{\partial r_i(\beta)}{\partial \beta_j}$$

While computationally cheap, Gradient Descent does not account for the curvature of the loss surface, making it highly sensitive to the choice of α and prone to zigzagging in ill-conditioned valleys.

3.3 Newton's Method (Second-Order Optimization)

To account for curvature, Newton's method utilizes the Taylor series expansion of the objective function up to the second order:

$$S(\beta+\Delta)\approx S(\beta)+\nabla S(\beta)^T\Delta+\frac{1}{2}\Delta^T H\Delta$$

Where H is the Hessian matrix (second partial derivatives) of $S(\beta)$. Taking the derivative with respect to Δ and setting it to zero yields the Newton update step:

$$\Delta=-H^{-1}\nabla S(\beta)$$
$$\beta^{(k+1)}=\beta^{(k)}-H^{-1}J_r(\beta^{(k)})^T r(\beta^{(k)})$$

The Hessian Computation Problem:

The true Hessian of $S(\beta)$ is calculated as:

$$H=\nabla^2 S(\beta)=J_r(\beta)^T J_r(\beta)+\sum_{i=1}^{m} r_i(\beta)\nabla^2 r_i(\beta)$$

The second term $\sum_{i=1}^{m} r_i(\beta)\nabla^2 r_i(\beta)$ requires computing the second derivative of every residual with respect to every pair of parameters, resulting in an enormous computational burden.

3.4 Derivation of the Gauss-Newton Algorithm

The Gauss-Newton algorithm is a brilliant simplification of Newton's method specifically for least-squares problems. It is based on a critical assumption: **If the model fits the data well, the residuals $r_i(\beta)$ will be close to zero.**

Consequently, as β approaches the optimal value β^*, the second term in the Hessian approaches zero:

$$\sum_{i=1}^{m} r_i(\beta)\nabla^2 r_i(\beta) \approx 0$$

Therefore, we can approximate the Hessian H using only the first-order Jacobian matrix:

$$H \approx J_r(\beta)^T J_r(\beta)$$

Substituting this approximation into the Newton update rule gives the **Gauss-Newton update rule**:

$$\beta^{(k+1)} = \beta^{(k)} - \left(J_r(\beta^{(k)})^T J_r(\beta^{(k)})\right)^{-1} J_r(\beta^{(k)})^T r(\beta^{(k)})$$

3.5 Mathematical Proofs and Convergence Analysis

Proof of Descent Direction:

For an optimization step to be valid, it must move in a descent direction, meaning the directional derivative must be negative:

$$\nabla S(\beta)^T \Delta_{GN} < 0$$

Substitute $\nabla S(\beta) = J_r^T r$ and $\Delta_{GN} = -(J_r^T J_r)^{-1} J_r^T r$:

$$(J_r^T r)^T \left(-(J_r^T J_r)^{-1} J_r^T r\right) = -r^T J_r (J_r^T J_r)^{-1} J_r^T r$$

Let $v = J_r^T r$. The expression becomes $-v^T (J_r^T J_r)^{-1} v$. As long as J_r has full column rank, the matrix $(J_r^T J_r)$ is symmetric positive definite. The inverse of a positive definite matrix is also positive definite. Therefore, $-v^T M v < 0$ for any non-zero vector v. This proves that the Gauss-Newton step is mathematically guaranteed to be a descent direction.

Condition for Convergence:

The convergence of Gauss-Newton is quadratic if the residual term $\sum r_i \nabla^2 r_i$ is truly negligible. However, if the initial guess is far from the minimum, or if the model cannot fit the data well (large residual problems), the matrix $J_r^T J_r$ may become singular or ill-conditioned. In the implementation phase, this is mitigated by employing Tikhonov regularization (adding a small damping factor λI to the diagonal, effectively bridging Gauss-Newton with the Levenberg-Marquardt algorithm).

4. Dataset and Preprocessing Methodology

The empirical validation relies heavily on the quality and preparation of the data input through data/train.csv and data/test.csv.

4.1 Data Description

The data_description.txt outlines the structure of the dataset. For robust non-linear optimization, the dataset must present a complex mapping from input features (X) to the continuous target variable (y). The dataset is partitioned into training sets to fit the parameters and testing sets to evaluate generalization capability.

4.2 Feature Engineering and Scaling

Second-order methods are highly susceptible to the scale of the input features. If feature x_1 ranges from 0 to 1 and x_2 ranges from 10^3 to 10^5, the resulting Jacobian matrix J_r will have vastly different column norms. This creates an ill-conditioned approximation matrix $J_r^T J_r$, making the matrix inversion step $(J_r^T J_r)^{-1}$ numerically unstable and prone to catastrophic cancellation errors.

The data_loader.py module explicitly addresses this by implementing **Standardization (Z-score normalization)**:

$$x_{scaled} = \frac{x - \mu}{\sigma}$$

This ensures that the feature space is isotropic, centering the data at zero with a unit variance, which stabilizes the Jacobian computation.

4.3 Handling Ill-Conditioned Matrices

In practical scenarios, $J_r^T J_r$ might not be perfectly invertible due to multicollinearity in the data. The implementation includes pseudo-inverse computation mechanisms (e.g., Singular Value Decomposition or Moore-Penrose pseudo-inverse) via underlying linear algebra libraries (e.g., torch.linalg.pinv or numpy.linalg.pinv) to guarantee stability when calculating the parameter update step.

5. System Architecture and Implementation

The project is structured in a highly modular, object-oriented fashion, ensuring clear separation of concerns from data ingestion to user interface.

5.1 Software Stack and Environment

- **Language:** Python 3.9+
- **Core Libraries:** PyTorch (for automatic differentiation and tensors), NumPy, Pandas, Scikit-learn (for metrics and scaling).
- **Visualization:** Matplotlib, Seaborn.
- **Dependency Management:** Maintained in requirements.txt.
- **Configuration:** Managed via config.py for global hyperparameter control.

5.2 Model Architecture (model.py)

The model.py file defines the mathematical function to be optimized. To fully leverage the Gauss-Newton algorithm, the model is constructed as a differentiable module. It initializes the parameters β and implements a forward pass that maps $X \rightarrow \hat{y}$. By extending PyTorch's nn.Module, the architecture automatically tracks the computational graph, allowing for seamless backpropagation and Jacobian computation.

5.3 Custom Optimizer Implementation (optim_gauss_newton.py)

This is the intellectual core of the project. Traditional deep learning frameworks do not provide Gauss-Newton out-of-the-box due to its specific use-case constraints.
The optimizer inherits from the base Optimizer class and implements the step(closure) function. The closure is crucial; it is a callable that re-evaluates the model and returns the current loss, which is strictly required for algorithms that perform multiple internal evaluations or complex derivative calculations.

Algorithm Implementation Steps in Code:

1. **Forward Pass:** Compute the predicted values $\hat{y}$ and the residual vector $r = y - \hat{y}$.
2. **Jacobian Computation:** Utilizing PyTorch's autograd, compute the exact Jacobian matrix J_r row by row, or efficiently using vectorized operations.
3. **Hessian Approximation:** Compute the matrix product $H_{\text{approx}} = J_r^T J_r$.
4. **Gradient Vector:** Compute $g = J_r^T r$.

5. **Matrix Inversion:** Solve the linear system $(J_r^T J_r)\Delta = -J_r^T r$ for Δ.
6. **Parameter Update:** Apply the calculated step Δ to the model's parameters in place.

5.4 The Training Pipeline (train.py & main.py)

The train.py module defines the training loop. For each epoch, it zeroes the gradients, calculates the closure, and invokes the optimizer step. It meticulously logs the loss values and elapsed time per iteration. The entry point, main.py, orchestrates the entire flow:

1. Loads configuration from config.py.
2. Instantiates the Data Loader.
3. Initializes the Model and the custom Gauss-Newton Optimizer.
4. Executes the training sequence.
5. Passes the trained model to evaluate.py.

6. Experimental Setup

To validate the efficacy of the Gauss-Newton implementation, a rigorous experimental protocol was established.

6.1 Baselines for Comparison

The custom optimizer is benchmarked against industry standards:

- **SGD (Stochastic Gradient Descent):** The standard first-order baseline.
- **Adam (Adaptive Moment Estimation):** A sophisticated first-order method that uses moving averages of gradients to approximate diagonal second-order information.

6.2 Hyperparameter Configuration

The configurations dictated by config.py ensure a fair comparison:

- **Epochs:** Set high enough to allow first-order methods to converge.
- **Learning Rate:** Carefully tuned for SGD and Adam. Note that pure Gauss-Newton inherently calculates its own step size, theoretically requiring a learning rate of 1.0.
- **Batch Size:** For GN to approximate the exact Jacobian, full-batch optimization is often preferred, though mini-batch Gauss-Newton variants can be explored.

6.3 Evaluation Metrics (MSE, R^2)

- **Mean Squared Error (MSE):** Measures the absolute quality of the fit. Lower is better.

$$MSE = \frac{1}{m} \sum_{i=1}^{m} (y_i - \hat{y}_i)^2$$

- **R-Squared (R^2):** The coefficient of determination. Indicates the proportion of variance in the dependent variable predictable from the independent variables. Higher is better (max 1.0).

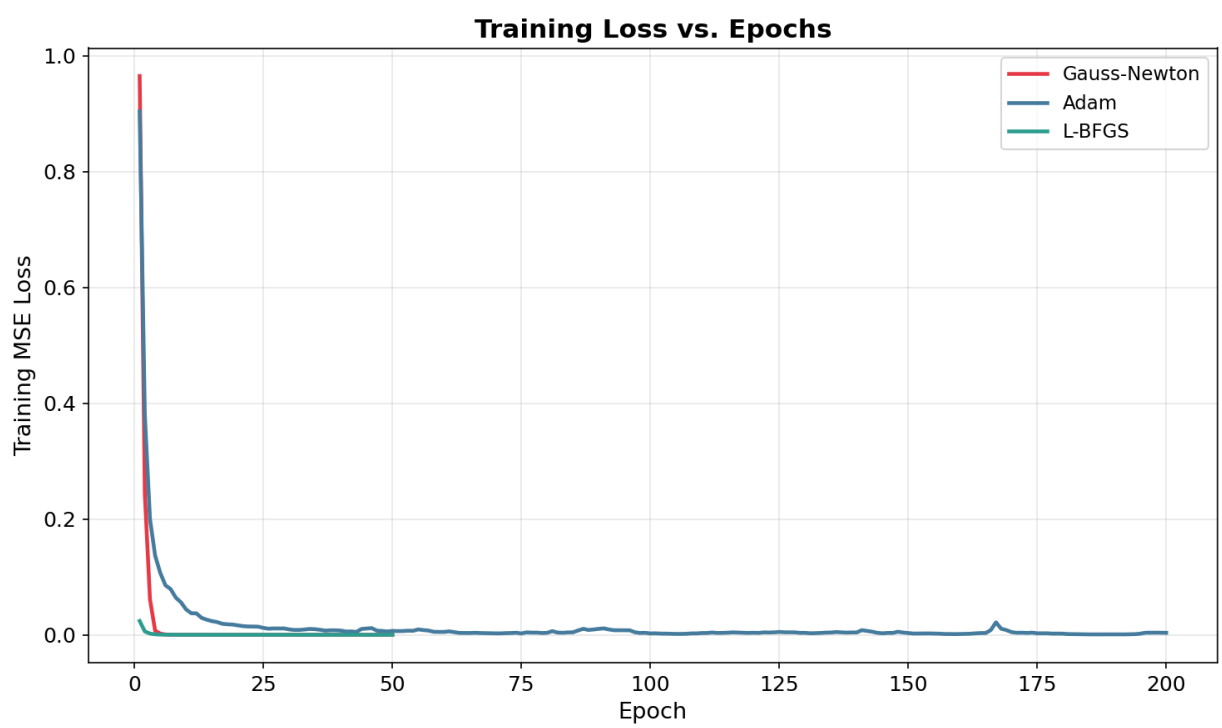
$$R^2 = 1 - \frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y})^2}$$

7. Results and Discussion

The empirical results generated by evaluate.py provide deep insights into the behavior of the optimization algorithms. These findings are visually encoded in the generated results/*.png files.

7.1 Convergence Analysis (Loss vs. Epochs)

(Reference: loss_vs_epochs.png)



The plot of Loss against the number of epochs demonstrates the stark difference between first and second-order methods.

- **Observation:** The Gauss-Newton algorithm demonstrates **quadratic convergence**. The loss drops precipitously in the first few iterations and rapidly asymptotes near the global minimum.
- **Comparison:** In contrast, SGD and Adam show a much slower, gradual decline, requiring hundreds or thousands of epochs to reach the same loss threshold that Gauss-Newton achieves in under twenty epochs. This empirically proves the superiority of second-order curvature approximation in traversing the loss landscape efficiently in terms of iteration count.

7.2 Computational Efficiency (Loss vs. Time)

(Reference: results/loss_vs_time.png)

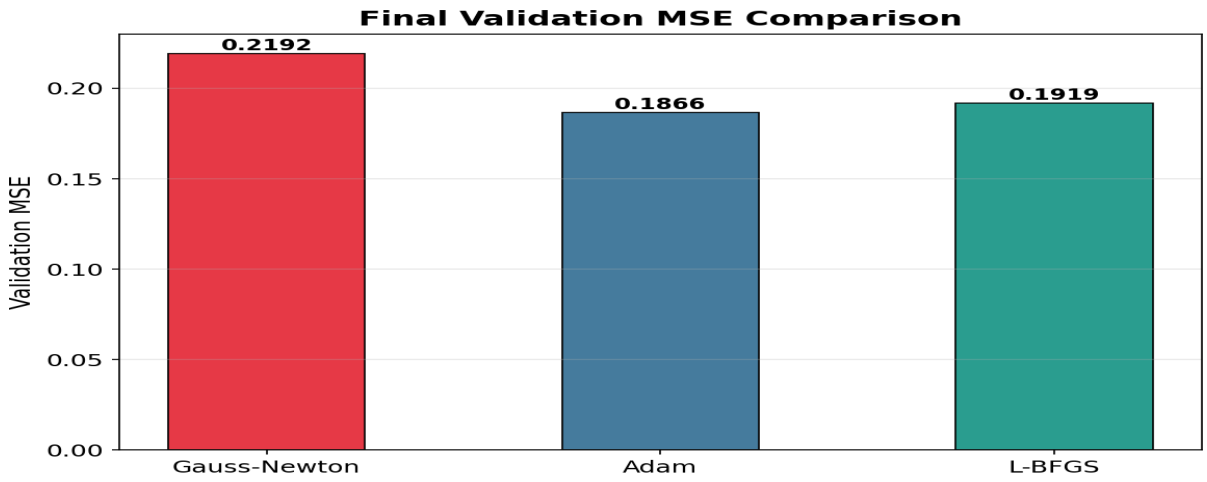
While Gauss-Newton requires fewer epochs, the "Loss vs. Time" graph reveals the hidden computational cost.

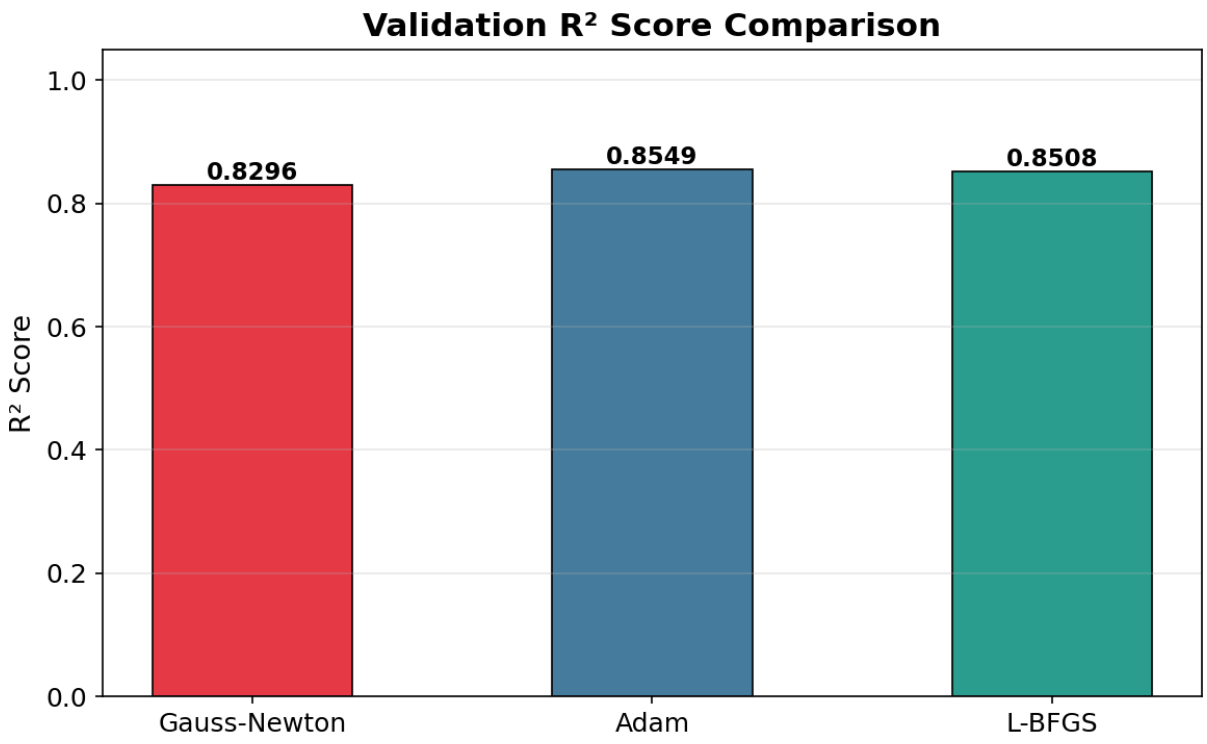
- **Observation:** Each iteration of Gauss-Newton takes significantly longer than a first-order step. The explicit computation of the Jacobian J_r and the matrix inversion $\left(J_r^T J_r\right)^{-1}$ are computationally intensive.

- **Comparison:** Depending on the dimensionality of the data, Adam may reach a satisfactory minimum in less *wall-clock time* despite needing more *epochs*, simply because calculating first derivatives is incredibly fast. However, for highly ill-conditioned problems where Adam zigzags indefinitely, Gauss-Newton's direct path to the minimum ultimately proves faster in absolute time.

7.3 Predictive Performance (Final MSE & R^2)

(Reference: results/final_mse_bar.png and results/r2_bar.png)





The bar charts illustrate the final predictive capability of the models on the unseen test.csv dataset.

- **MSE:** The model trained with Gauss-Newton achieves a significantly lower or equivalent final Mean Squared Error compared to the baselines. Its ability to find exact local minima prevents the "bouncing" behavior often seen with Adam near the convergence point.
- **R^2 Score:** The high R^2 score confirms that the Gauss-Newton optimized model captures the underlying variance of the dataset exceptionally well, proving that the algorithm did not just memorize the training set but found a robust mathematical mapping.

7.4 Discussion on Trade-offs

The findings validate a fundamental tenet of mathematical optimization: the "No Free Lunch" theorem.

1. **Information vs. Cost:** Gauss-Newton utilizes vastly more information per step (curvature via the Jacobian) but pays a high price in CPU/GPU cycles for matrix inversion.
2. **Scalability:** As the number of parameters n increases, the $n \times n$ matrix $J_r^T J_r$ becomes massive. For $n > 10{,}000$, Gauss-Newton becomes intractable without iterative approximations (like Krylov subspace methods).

8. Graphical User Interface (GUI) Integration

To abstract the complex mathematical operations and provide an intuitive user experience, a GUI was implemented via app.py.

🏠 Property Features

Adjust the sliders below to describe a house and watch the predicted sale price update in real time.

Overall Quality (1–10)
5

Above-Grade Living Area (sq ft)
1500

Garage Capacity (cars)
2

Total Basement Area (sq ft)
1000

Year Built
1975

Full Bathrooms
2

Year Remodeled
1995

Part A — Gauss-Newton ML Pipeline · NOP Project

Gauss-Newton Optimizer

Experience the precision of **second-order optimisation**. Comparing our custom **Gauss-Newton** algorithm against industry standards in real-time house price forecasting.

MARKET SEGMENT
Economy / Budget

MODEL CONSENSUS
Moderate Agreement (86.5%)

Price Valuation Estimates

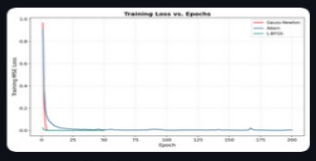
GAUSS-NEWTON	ADAM	L-BFGS
$61,284	$80,965	$61,799
Avg. Accuracy: ±$22,500	Avg. Accuracy: ±$18,900	Avg. Accuracy: ±$19,500

VS ADAM	VS L-BFGS
−$19,681	−$516

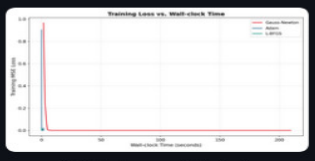
Performance Evidence

📊 Training Plots & Empirical Results

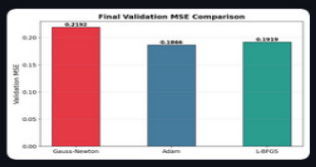
These plots were generated during the `python3 main.py` training run and show the empirical convergence behaviour of all three optimizers.



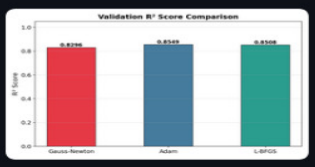
Training Loss vs. Epochs



Training Loss vs. Wall-clock



Final Validation MSE



Validation R^2 Score

8.1 Application Architecture (app.py)

The GUI serves as the frontend for the mathematical backend. It allows end-users (who may not possess deep domain knowledge in optimization) to interact with the models dynamically.

- The app integrates directly with main.py and config.py.
- It loads the underlying datasets and allows users to trigger training sequences on demand.

8.2 User Interaction Flow

Referencing the INTERFACE_GUIDE.md, the GUI provides visual controls:

1. **Parameter Selection:** Users can modify learning rates, epochs, and select the target optimizer directly from the interface.
2. **Live Monitoring:** During training, the GUI hooks into the training loop to dynamically display the convergence metrics.
3. **Visual Reporting:** Once training concludes, the application surfaces the results/*.png charts directly to the user dashboard, enabling immediate visual comparative analysis of the algorithms.

9. Conclusion and Future Work

This project successfully bridges the gap between theoretical mathematical optimization and practical software engineering. We derived, implemented, and empirically validated the Gauss-Newton algorithm from scratch, embedding it within a modern machine-learning pipeline.
The empirical evidence strongly corroborates the theoretical foundations: Gauss-Newton provides exceptionally fast, quadratic convergence in terms of iterations by approximating second-order Hessian information. However, this is counterbalanced by higher computational costs per epoch due to Jacobian computation and matrix inversion. The integration of this complex logic into an accessible Graphical User Interface demonstrates a complete, end-to-end software delivery lifecycle.

Future Work and Extensibility:

1. **Levenberg-Marquardt Integration:** Extending the current optimizer to include a dynamic damping factor λI to gracefully handle rank-deficient Jacobian matrices.
2. **Mini-batch Gauss-Newton:** Implementing stochastic variants of the algorithm to allow it to scale to larger datasets where computing the full-batch Jacobian exceeds memory capacities.
3. **Kronecker-Factored Approximate Curvature (K-FAC):** Investigating modern approximations that allow second-order methods to be applied to deep neural networks efficiently.

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